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Xiang(Shawn) Li, Ph.D.

Interests

Agentic AI & Computational Genomics

Education

Ph.D., George Washington University, Physics (**Honor: Parke Prize**), 2016 ~ 2021
Thesis Summary Study of Transcriptional and Post-transcriptional Regulation in Immune Systems
B.S., Huazhong University of Science & Technology, Applied Physics, (**Honor**) 2011 ~ 2015

Work Experiences

PSC, Carnegie Mellon University Senior Staff — Pittsburgh PA, 2024 ~ Present

- **Partnership** Human BioMolecular Atlas Program: Lead the collaboration between TACC & PSU Galaxy, By building AI-enabled research infrastructure and workflows(HuBMAP) under Common Fund Data Ecosystem (CFDE)
- **Platform Built**: AI-assisted data research + multi-agent analytics infrastructure.
- **Support**: Contributors to Open AI projects: Harbor, LangChain & Internal Evaluation
- **Lead** a R03 grant: multi-agent platform(MCP server + LangGraph/deepagents architecture), enabling enabling distributed scientific querying, evidence retrieval, and bioinformatic analysis execution.

Johnson & Johnson Scientist III, RWE, End due to working VISA(H1B). — Titusville NJ, 2024

- R&D in Real World Evidence, Patient profiling from medical history (EHR, EMR)
- Dev a toolset from scratch for patient profile, alignment, visualization and analysis(built a fully connected network for 52K patients).

GentiBio, Senior Scientist, Bioinformatics Lead, End due to Reorg — Boston MA, 2022 ~ 2024

- **Translational Research**: GNTI-122 IND package (Safety assessment for Gene Editing).
  - Build SNIP-aware (gnomAD v4.0) insilico Off-target analysis for Cas9 editing
  - On/Off-Target analysis protocol: in-Silico ⇒GUIDESeq ⇒rhAMPSeq.
  - Build a full-stack web app (aws) for automation. Similar to CRISPResso2 Web
- **Early Discovery**, Scientist II: Cellular Profiling Platform
  - Research: single cell multi-omics integration analysis
  - Dev: AWS VC-backed start-up package and build bioinformatics infrastructure

Sana Biotechnology Scientist, Computational Biology, — Seattle WA, 2021 ~ 2022

- **Corporate team**, research on target discovery & dev pipelines automation.
- **iPSC Master Cell Bank**, Cell Characterization & Safety Assessment

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GWU, Department of Physics, — Washington, DC, 2017 ~ 2021
Ph.D, (Superviser: *Prof. W. Peng* , Collaborator: *Prof. H. Xue*)

- *Nature Com:* Develop analysis framework (python-base for data representation, visualization and ML)
- Integration of Multi-omics: scRNAseq(10X), RNAseq and DNase_seq, ATAC_seq, ChIPseq, Cut&Run, HiC, MiCroC, HiChIP and etc.
- Chromatin structure and function: Develop a tool (HiCHub, which solved the problem on identifying a mixed range of altered genome structure(from looping to larger than TAD). This tool outperforms similar software in HiC data analysis with a significantly increase in the computational efficiency.

- Nucleic Acids Research: mRNA stability Mathematical Modeling and data analysis.
- Study regulatory roles of Tcf1/Lef1 in $CD8^+$ T cells in immunue response. (JEM, 2019, 2018)
- Impact upon deficiency of miRNA-34b/c during somatic reprogramming progress.
- Organize Journal Club and give presentation once per month.
- HPC management. (Linux Admin, data storage &sharing& backup, computation dependencies setup, jupyter notebook & Rstudio server and on-call for tech emergency.
- Build analysis pipeline, dependencies and visualization tools on a cloud platform. (AWS, Linux ssh).

Team Lead, NLP Course Project
George Washington University — Washington, DC, 2019

- Natural language processing, use different models for user profiling task in given posts from twitter.
- Write model to implement different language models, Hidden Markov model and bayesian classifier.

Joint Leader of Student Innovation Group (Adviser: B. Yu)
Qiming College of Huazhong University of Science & Technology — Boston, 2014

- Built a team for 2014 *BIOMOD, Harvard*, Cambridge, Massachusetts.
- Invested nano-scale transportation system based with magnetic control device.
- Conducted the final presentation on behalf of our team.
- For more detail, please visit our website: *BIOMOD 2014: HUST*

Teaching

- Graduate TA. General Physics (with: *X. Qiu* / *G. Younes* / *S. Guiriec*) — 2016 ~ 2018
- Instructor. Astrophysics (with: *Kalvir Dhuga*) — 2019

Membership & Reviewer

- Phenomics, Youth Editorial Board Members — 2024 ~ Present
- Society of Artificial Intelligence Research SAIR — 2023 ~ Present
- Machine Learning for Genomics MLGenX — 2024

Honors and Awards

- Best Posters Award, Common Fund Data Ecosystem (CFDE) Spring 2026 Conference
- The Parke Prize, awarded for excellence in theoretical biophysics
- Scholarship in Department of Biostatistics, University of Washington, $1,950
- Second Prize of Poster in 3rd International Ocean Sciences Summer School. $800
- Project Awards: Silver; International Biomolecular Design Competition 2014, *BIOMOD, Harvard*
- Fellowship $12000 total, Undergraduate Fellowship from College of Life Science & Technology,

Publications & Patents & Presentations

- Inhibition of LARP4-mediated quiescence exit of naive CD4+ T cells ameliorates autoimmune and allergic diseases Jian Zhou, Di Yang, Chao Han, Hui Dong, Shufeng Wang, **Xiang Li**, ..., Yi Tian†., Nature Biomedical Engineering (2025)
- HuBMAP Data Portal: A Resource for Multi-Modal Spatial and Single-Cell Data of Healthy Human Tissues Jian Zhou, Di Yang, Chao Han, Hui Dong, Shufeng Wang, **Xiang Li**, ..., Yi Tian†., Nature Communications (NCOMMS-25-81343)
- HiCHub: A Network-Based Approach to Identify Domains of Differential Interactions from 3D Genome Data **Xiang Li**†, ..., Hai-Hui Xue, Weiqun Peng., Bioinformatics (BIOINF-2025-0207)
- Lysine methyltransferase Kmt2d regulates naive CD8+ T cell activation-induced survival Jaekwan Kim, T.N, ..., Xiang Li, ..., Nan-ping Weng., Frontiers in Immunology (2023)
- Tcf1 CTCF cooperativity shapes genomic architecture to promote CD8+ T cell homeostasis Q. S†, ..., **Xiang Li**,..., Hai-Hui Xue., Nature Immunology (2022)

- Tcf1 and Lef1 maintain CD8+ T cell identity by organizing genomic architecture Q. S†, **Xiang Li**†,..., Hai-Hui Xue., Nature Communications (2021)
- **Selected Oral Presentation**, 2021 Network Biology, Cold Spring Harbour Laboratory Network analysis reveal the critical role of TCF1/LEF1, Speaker: **Xiang Li**
- Sepsis leads to lasting changes in phenotype and function of memory CD8 T cells Isaac J. Jensen†, **Xiang Li**†,..., Weiqun Peng and Vladimir P. Badovinac., 2021 eLife
- Arid1a-Plagl1-Hh signaling is indispensable for differentiation-associated cell cycle arrest of tooth root progenitors Yang Chai, ..., **Li Xiang**, Xinquan Jiang. Cell Reports (2021).
- Transcriptome-wide stability analysis uncovers LARP4-mediated NFB1 mRNA stabilization during T cell activation. Tian Y.†, Zeng Z.†, **Li Xiang**†, ..., Peng W. and Zhu J. Nucleic Acids Research (2020).
- Integrated analysis of carbon dioxide and oxygen concentrations as a quality control of ocean float data Yingxu Wu, D. Bakker, E. Achterberg, ..., **Xiang Li**, ... Communications Earth Environment, Accepted. (Mar, 2022)
- Tcf1 and Lef1 transcription factors are required for the immunosuppressive function of regulatory T cells Xing S., Gai K., **Li Xiang**,..., Peng W. and Xue H., J Exp Med (2019).
- Tle corepressors are differentially partitioned to instruct CD8+ T cell lineage choice and identity Xing S., ..., **Xiang Li**,..., Peng W. and Xue H., J Exp Med (2018).

Work extensively used HiCHub

- Tcf1 CTCF cooperativity shapes genomic architecture to promote CD8+ T cell homeostasis Nature Immunology (2022)
- CTCF mediates CD8+ effector differentiation through dynamic redistribution and genomic reorganization JEM (2023)